

6.200 CHEATSHEET

Selinna Lin

Core Laws + Conventions

- charge: $Q = I \cdot t$
- KCL: sum of currents leaving or entering a node = 0
- KVL: sum of voltages around a closed loop = 0
- Ohm's Law: $V = IR$
- Passive Sign Convention: $p = vi$ (current enters terminal, consumes power)
- Voltage Divider: Resistors in series $V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$
- Current Divider: Resistors in parallel $I_1 = I_{total} \cdot \frac{R_2}{R_1 + R_2}$

Capacitors & Inductors

$Z_C = \frac{1}{j\omega C}$
 $Z_L = j\omega L$

$E_C = \frac{1}{2} CV^2$
 $E_L = \frac{1}{2} Li^2$

$C \frac{dV}{dt} = I$
 $L \frac{di}{dt} = V$

$C_{eq} = \sum C_n$
 $i_{eq} = \sum i_n$ } parallel

$\frac{1}{C_{eq}} = \sum \frac{1}{C_n}$
 $i_{eq} = \sum i_n$ } series

LC Circuit

$\omega = \frac{1}{\sqrt{LC}}$ Natural frequency.

$\frac{V_{amp}}{Lamp} = \frac{1}{\sqrt{LC}}$ characteristic impedance.

$i(t) = V_0 \sqrt{\frac{C}{L}} \sin(\omega t)$

$V_C(t) = V_0 \cos(\omega t)$

$V_L(t) = -V_0 \cos(\omega t)$

Converting to Decibels

x	$x_{dB} = 20 \log_{10}(x)$
1/100	-40
1/10	-20
1/√2	-3
1	0
√2	3
10	20
100	40

$0.7 \approx \frac{1}{\sqrt{2}}$

Frequency Response

capacitor $\frac{1}{s}$

inductor s

low frequency high frequency

Real-valued poles: # of times slope ↓ by 20 dB/dec (mag); # of times phase ↓ by $\pi/2$

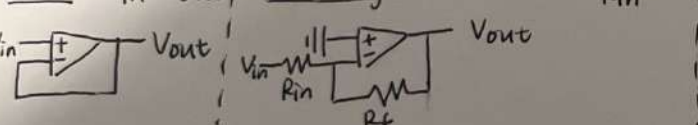
Real-valued zeroes: # of times slope ↑ by 20 dB/dec (mag); # of times phase ↑ by $\pi/2$

flat midband gain: Resistor Ratio.

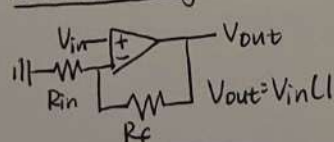
Operational Amplifiers (Op-amps)

Ideal op-amp Assumption: Net negative feedback. $V_+ = V_-$ and $i_+ = i_- = 0$.

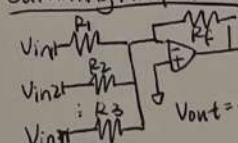
Buffer $V_{in} = V_{out}$; Inverting Amplifier $V_o = -\frac{R_f}{R_{in}} V_{in}$



Non-Inverting Amplifier



Summing Amplifier



Conductance-Weighted Average

Two Thevenin's in parallel: $V_o = \frac{V_1/R_1 + V_2/R_2}{1/R_1 + 1/R_2}$ Assuming polarities at the same terminals.

$\sin(\omega t) = \cos(\omega t - \frac{\pi}{2})$
 $\cos(\omega t) = \sin(\omega t + \frac{\pi}{2})$ } delay cos by $\frac{\pi}{2}$ shifting cos to the left.

Transfer Functions + Completeness

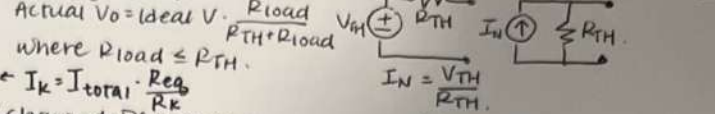
- General Strat:
1. Replace blocks with impedances
2. $H(s) = \frac{Z_{out}}{Z_{in}}$
3. Factor denominators → poles
4. Numerator → zeroes.

Step + Sinusoid Input:

$V_{out}(t) = V_{ss}(t) + \sum A_i e^{p_i t}$
 V_{ss} : Steady State Sinusoid
 p_i : poles of $H(s)$

Thevenin/Norton

V_{TH} : open-circuit the load, compute terminal voltage.
 R_{TH} : If no dependent sources, turn off sources: $R_{TH} = R_{eq}$. If dependent sources, apply test source: $R_{TH} = V_{test} / I_{test}$.



At the instant of switching ($t = 0^+$): V_C and i_L are continuous. $V_C(0^+) = V_C(0^-)$, $i_L(0^+) = i_L(0^-)$

Time-domain Behavior: (charged) DC steady-state $t \rightarrow \infty$

capacitor acts as an open (current = 0A)

Inductor acts as a short (voltage drop = 0)

First-Order: τ : time constant (1/3 of the way)

RL circuit: $i_L(t) = i_N + (i_L(t=0) - i_N)e^{-t/\tau}$ $\tau = \frac{L}{R}$

RC circuit: $V_C(t) = V_{TH} + (V_C(t=0) - V_{TH})e^{-t/\tau}$ $\tau = RC$

Second-Order RLC Circuits

$\omega_0 = \frac{1}{\sqrt{LC}}$
 $\alpha = \frac{R}{2L}$ (series)

$\omega_0 = \frac{1}{\sqrt{LC}}$
 $\alpha = \frac{1}{2RC}$ (parallel)

$Q = \frac{\omega_0}{\alpha} = \frac{\sqrt{L}}{R\sqrt{C}}$

$\alpha > \omega_0$: overdamped $Q < \frac{1}{2}$

$\alpha = \omega_0$: critically damped $Q = \frac{1}{2}$

$\alpha < \omega_0$: underdamped $Q > \frac{1}{2}$

$s_{1,2} = -\alpha \pm j\omega_d$

$A_1 e^{s_1 t} + A_2 e^{s_2 t}$

$(A + Bt)e^{-\alpha t}$

$e^{-\alpha t} (A \cos(\omega_d t) + B \sin(\omega_d t))$

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

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